

## Approximation of areas



1 Use the trapezoidal rule to find an approximation for each integral:

a  $\int_2^5 3x^2 dx$

b  $\int_0^2 (2x^3 - 1) dx$

c  $\int_2^4 dx$

2 Find an approximation to  $\int_0^1 x^3 dx$  using the trapezoidal rule with:

a 1 subinterval

b 2 subintervals

3 Given the table of values, find the approximate value of each definite integral:

a  $\int_0^8 f(t) dt$

|        |   |   |    |   |   |
|--------|---|---|----|---|---|
| $t$    | 0 | 2 | 4  | 6 | 8 |
| $f(t)$ | 3 | 7 | 11 | 9 | 3 |

b  $\int_0^{40} f(x) dx$

|        |     |     |     |     |     |
|--------|-----|-----|-----|-----|-----|
| $x$    | 0   | 10  | 20  | 30  | 40  |
| $f(x)$ | 350 | 410 | 435 | 450 | 460 |

c  $\int_{-4}^2 f(x) dx$

|        |    |    |    |    |    |    |   |
|--------|----|----|----|----|----|----|---|
| $x$    | -4 | -3 | -2 | -1 | 0  | 1  | 2 |
| $f(x)$ | 0  | 4  | 5  | 3  | 10 | 11 | 2 |

4 Use the trapezoidal rule with 2 subintervals to find an approximation for the following to three decimal places:

a  $\int_3^4 \ln 2x dx$

b  $\int_0^6 \frac{dx}{x+3}$

5 Find an approximation to the following integrals, rounding your answers to three decimal places:

a  $\int_2^3 e^{2x} dx$  using 3 subintervals

b  $\int_0^2 (4x^2 - 16) dx$  using 4 subintervals

c  $\int_0^1 \sqrt{2x} dx$  using 5 subintervals

d  $\int_1^3 \frac{dx}{x^3}$  using 4 subintervals

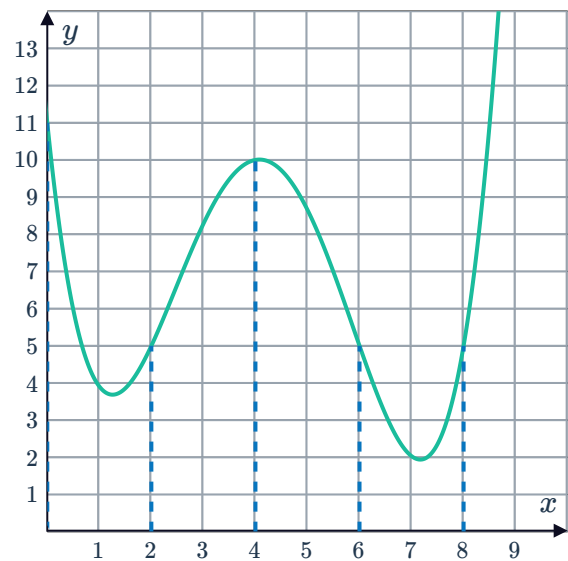
e  $\int_2^5 \frac{2}{3x-4} dx$  using 6 subintervals

Use one application of the trapezoidal rule to approximate  $\int_2^3 \frac{e^x}{x} dx$  to one decimal place.

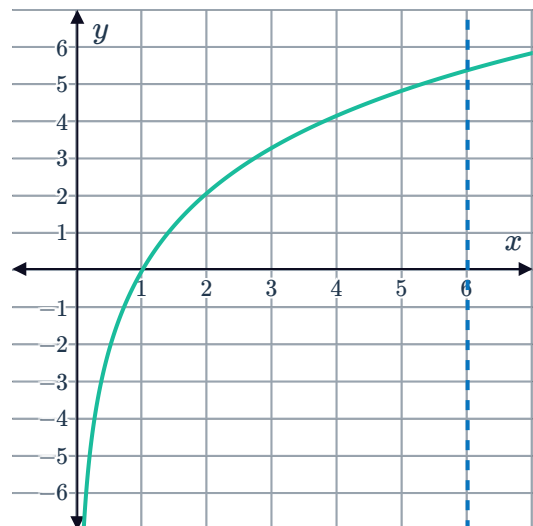
- 7 Use three applications of the trapezoidal rule to approximate  $\int_0^9 e^{-x^2} dx$  to one decimal place.
- 8 Approximate  $\int_0^8 8x dx$  by using four rectangles of equal width whose heights are the values of the function at the midpoint of each rectangle.
- 9 Approximate  $\int_1^5 \frac{1}{x} dx$  by using four rectangles of equal width whose heights are the values of the function at the right endpoint of each rectangle.

- 10 In the following graph, the interval  $[0, 8]$  is partitioned into four subintervals  $[0, 2]$ ,  $[2, 4]$ ,  $[4, 6]$ , and  $[6, 8]$ :

- a Approximate the area  $A$  using rectangles for each subinterval whose heights are equal to the function values of the left side of the subintervals.
- b Approximate the area  $A$  using rectangles for each subinterval whose heights are equal to the function values of the right side of the subintervals.



- 11 The function  $y = 3 \ln x$  has been graphed: Use two applications of the trapezoidal rule to approximate the area bound by the curve, the  $x$ -axis and  $x = 6$ . Round your answer to one decimal place.



12 Consider the function  $y = e^{x^2}$ .

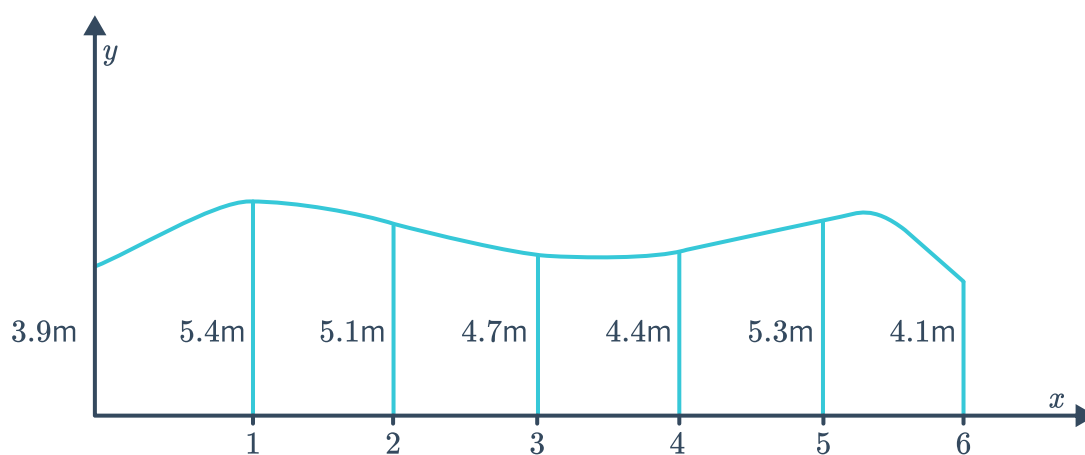


- a Determine  $y''$ .
- b Using two applications of the trapezoidal rule, approximate the area bound by the curve and the  $x$ -axis between  $x = 0$  and  $x = 2$ . Round your answer to one decimal place.
- c Is the approximation given by the trapezoidal rule an underestimate or an overestimate of the actual area? Explain your answer.

13 Consider the integral  $\int_1^5 (\ln x + 4) dx$ .

Approximate the area bounded by the curve, the  $x$ -axis,  $x = 1$  and  $x = 5$  using two applications of the trapezoidal rule. Round your answer to two decimal places.

14 Use the trapezoidal rule to find the approximate area of the irregular figure below:

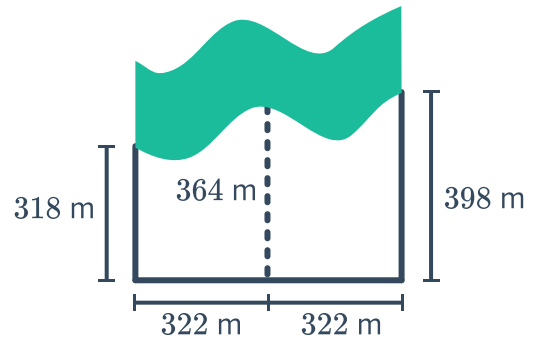




## Applications

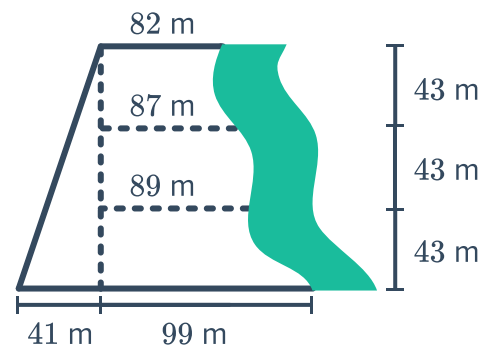
**15** The following piece of land has straight boundaries on the east, west and south borders and is bounded by a creek to the north. The land has been divided into two sections so we can use the trapezoidal rule to approximate the area:

- a** Find the approximate area of the piece of land by using two applications of the trapezoidal rule.
- b** During a heavy storm, 35.2 mm of rain fell. Find the volume of water that falls on this land. Round your answer to the nearest cubic metre.



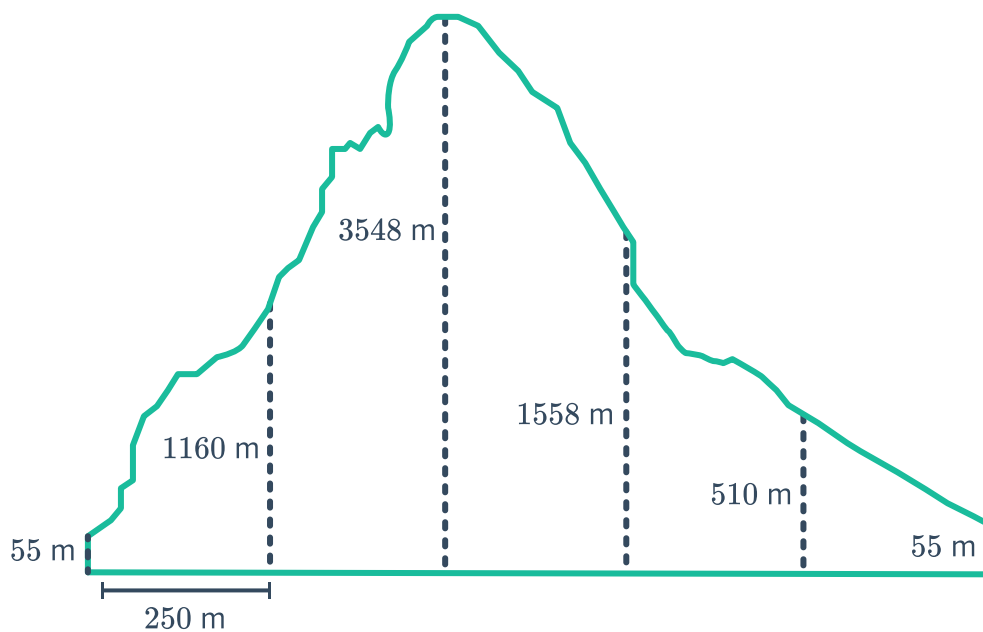
**16** A surveyor made the following diagram with measurements for a property she was mapping out. On the west side of the property is a river:

- a** Find the approximate total area of the property by using three applications of the trapezoidal rule.
- b** The average weekly rainfall is 34 mm. Calculate the total volume of water that falls on the land in cubic metres. Round your answer to two decimal places.



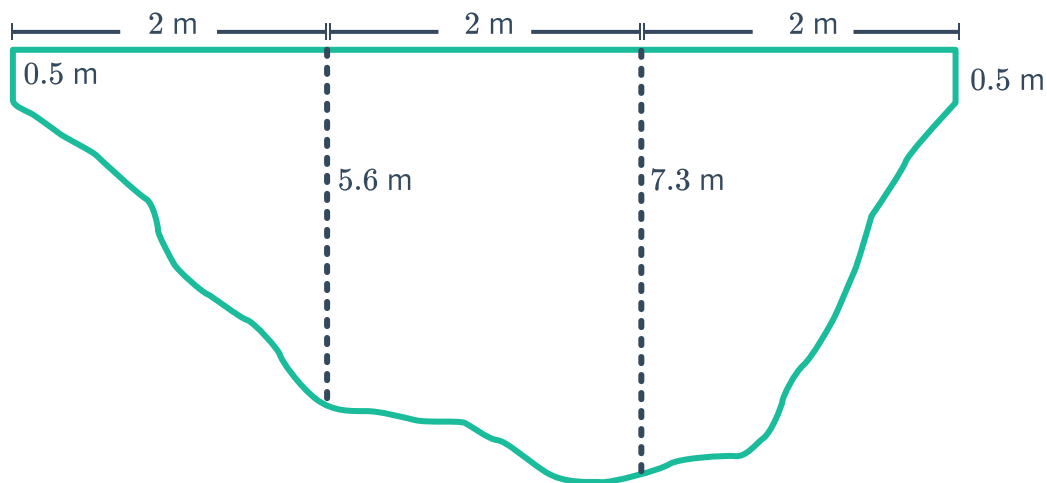
**17** A river has its depths marked out at equal intervals of 9 m. The depths are 0, 12, 14, 17, 5, and 0 m respectively. Find the approximate area of the cross section of the river.

- 18 The elevation values of a mountain are recorded at equal intervals of 250 m. The heights are shown in the diagram:



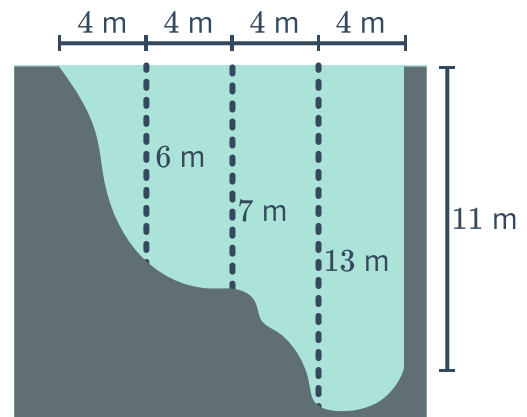
Find the approximate area of the cross section of the mountain.

- 19 The diagram shows the cross-section of a river. The depths of the river are marked at 2-metre intervals:

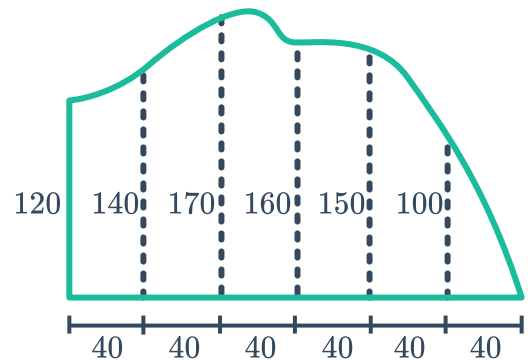


Using three applications of the trapezoidal rule, approximate the area of the cross-section of the river to one decimal place.

- 20 Use four applications of the trapezoidal rule to approximate the area of the cross-section of the following river:



- 21 The following shape has measurements given in metres. Use the trapezoidal rule to find the area in hectares.



- 22 A garden is 49 m long. At 7 m intervals, the width of the garden was given by the following measurements:

0 m, 2.9 m, 5.2 m, 6.6 m, 5.6 m, 4.3 m, 3 m, 2.5 m

Using the trapezoidal rule, approximate the area of the garden to two decimal places.